



NILAY TOSHNIWAL

Course : **B.E. (Hons.)**, Electronics and Communication Engineering, 2027

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CGPA : 7.18



ACADEMIC DETAILS

COURSE	INSTITUTE/COLLEGE	BOARD/UNIVERSITY	SCORE	YEAR
CLASS XII	Delhi Public School, Pune	CBSE	94.2 %	2023
CLASS X	Sarhad School	CBSE	94.4 %	2021

Subjects / Electives	Probability and Statistics, Machine Learning for Electronics, Digital Signal Processing, Signals and Systems, Control Systems, Data Structures, Computer Programming, Linear Algebra and Calculus, Natural Language Processing
Technical Proficiency	Python, SQL, C++, Pandas, NumPy, Scikit-Learn, PyTorch, Time Series Analysis, Statistical Modeling, Predictive Modeling, Root Cause Analysis, Docker, MLflow, ETL

INTERNSHIPS

Research and Development Engineering Intern, ElasticRun	Jun 2026 - Aug 2026
Built the bimanual robot platform end to end in ROS2, Gazebo and MuJoCo , with 7-DOF arm control through inverse kinematics and torque-mode PD control. Traced a robot arm's runaway oscillation to a damping term fighting the physics solver, dropping steady-state joint velocity from 7.27 to 0.076 rad/s. Audited a production sorter simulator against its specification, found seven engine-level defects, and proved each fix on a 1,341-case regression suite. Built the vision layer of the parcel line to catch damaged goods before dispatch, reaching 95% detection accuracy alongside label and barcode reading.	
Backend Developer, MyJobFactory.com	May 2025 - Jul 2025
Led the six-person backend sub-team. Built a semantic matching engine that ranks candidates against 500+ job descriptions, cutting the manual screening work a recruiter had to do by 40% . Turned it into Python pipelines over large structured and unstructured datasets, deployed as a live scoring service.	

COMPETITIONS

Winners - TATAVayu AI Studio Hackathon	Jul, 2026
Led a team of four to build a PCB inspection system that finds six fabrication defect types with no golden reference board, so a new design needs no setup. Every inspection ends in a pass, hold or scrap call with a repair cost, shipped as a live GPU service on Docker, S3 and MLflow with one-command rollback. Validated on board layouts the model had never seen in training, so the result was generalisation rather than a memorised set.	
Runners Up - APPCAIR JAW 2026 Hackathon	Aug, 2026
Rebuilt a hidden relational database out of 687 unstructured filings, resolving the same entity across sources that openly contradicted each other. Answered all 333 questions by running real queries instead of generating text, scoring 100/100 and holding it across 3,097 reworded questions.	

PROJECTS

Time-Series Volatility and Regime Research - Statistical Modelling	Aug 2025 - Dec 2025
Studied how volatility travels between equity and currency markets , to find the periods when risks stack up together instead of cancelling out. Modelled time-varying volatility and correlation with GARCH across 28 equities , isolating regimes where cross-asset risk amplifies rather than diversifies. Fixed five go/no-go thresholds before running anything, then ran an adversarial audit of my own engine that surfaced four real defects.	
Natural-Language Query Agent over Time-Series Data - LLM and SQL	Apr 2026 - Jul 2026
Built an agent answering plain-English questions about time-series data by writing its own SQL , so an analyst never has to know the schema. Computed return, volatility, drawdown and Sharpe over daily series for 29 tickers through guarded SQL against a normalised database . Scored 93.8% on a 48-question benchmark with independent ground truth.	
Rule Gated Autonomous Agent with Live Monitoring - Risk Analysis	Dec 2025 - May 2026
Deployed an agent that acts on live market data only when eleven numbered risk gates all pass, with circuit breakers that halt it when conditions break. The gates cut backtested maximum drawdown from 4.38% to 1.39%, with every decision written to a Merkle-tree-audited log . Gated entry on a pre-registered volatility risk premium of 3.68 points over 1,741 observations, with autocorrelation-robust errors at $t = 4.74$.	
Edge Inference on a Custom RISC-V Processor - Embedded ML	Dec 2025 - Present
Put a trained neural network onto a processor I designed myself, to find what it really costs to run a model next to the sensor instead of in the cloud. Quantised it to INT8, verified bit-identical against a Python reference, then added a hardware accelerator cutting inference from 139,813 to 4,241 cycles. A 32.8x speedup for a 0.03 percentage-point accuracy cost, measured on hardware rather than estimated.	
TRABSA: Hybrid Transformer-BiLSTM Sentiment Classifier - Deep Learning	Jan 2026 - May 2026
Built a classifier for seven mental-health categories across 52,681 posts, to test whether a custom architecture beats a well-tuned simple baseline. Designed the architecture myself, RoBERTa into an 8-head attention layer into a BiLSTM , reaching 81.3% accuracy against a 74.2% TF-IDF baseline . Compared it against two independent baselines and mapped where it fails, isolating the pair of classes the task cannot separate.	

POSITION OF RESPONSIBILITY

Core Member - Society of Automotive Engineers (SAE), BITS Pilani	Aug 2024 - Aug 2025
Led subsystem design and hardware integration for 2 competitive builds and resolved twelve critical electrical faults during pre-competition bring-up, isolating sensor and power-supply failures with bench supply, DMM and thermal measurements.	